



FUSION
SOFTWARE INSTITUTE

SOFTWARE TESTING



www.fusion-institute.com

About Fusion

Welcome to Fusion Software Training Institute, where innovation meets excellence. Our mission is to empower individuals and professionals with cutting-edge skills in software development and technology. With a curriculum designed by industry experts, we offer hands-on training that prepares you for real-world challenges. Our dedicated instructors provide personalized mentorship, ensuring you gain both technical and soft skills crucial for career success. Join our vibrant community of tech enthusiasts and leverage our state-of-the-art facilities to unlock your potential and transform your career. Fusion is your gateway to excelling in the dynamic tech industry. Together, we shape the future of technology.

Certifications



Course Introduction

In today's fast-paced digital world, delivering reliable, high-quality software is crucial to business success across industries—from fintech and e-commerce to healthcare and cloud services. At Fusion Software Institute, our industry-aligned Software Testing program empowers you with the hands-on skills needed to ensure quality, functionality, and performance at every stage of the development lifecycle.

Whether you're manually testing business workflows, validating APIs with Postman, or automating regression testing with Selenium, our course covers it all. You'll build a strong foundation in SDLC, defect lifecycle, test case design, and master advanced testing frameworks like TestNG, POM, BDD-Cucumber, and tools like JIRA.

Learn to perform real-time API testing using Postman, including authentication, CRUD operations, response validations, and common troubleshooting. Our curriculum also includes database testing using MySQL Server, core Java fundamentals, and Unix/Linux commands essential for testers in agile environments.

Gain deep exposure to automation testing, recovery scenarios, debugging, and real-world project simulations. With expert-led sessions, case-based learning, and live project implementation, you'll be prepared for in-demand roles such as QA Engineer, Test Automation Engineer, API Tester, and Test Analyst.

Step confidently into the tech world where precision meets performance—only at Fusion Software Institute.

Tools Covered



TestNG

cucumber



Course Content

SUMMARY

- Manual Testing
- Core Java Using Eclipse
- API Testing With Postman Tool
- Database Testing Using MySQLServer 2014
- Automation Testing Using Selenium Web Driver
- TestNG Framework
- POM Framework
- BDD Cucumber Framework
- Jira
- Linux/Unix Commands
- 2 Industry Level Live Projects

INTRODUCTION

- Introduction To Testing
- What Is Testing?
- Importance Of Testing
- Common Problem In Testing
- What Is Software Quality?
- Objective Of Testing

DIFFERENT ISSUES

- What Is Error?
- What Is Fault?
- What Is Bug?
- What Is Defect?
- What Is Mistake?
- What Is Flaw?

SDLC

› What Is SDLC?

› Phases In SDLC

- Requirement Gathering
- Planning
- Design
- Development
- Testing
- Deployment
- Maintenance & Hypercare

SDLC MODELS

› Waterfall Model

› Spiral Model

› V Model

› Agile Testing Implementation

› Agile SCRUM

KINDS OF TESTING

- › Unconventional Testing
- › Conventional Testing

METHODS OF TESTING

- › Black Box Testing
- › White Box Testing
- › Gray Box Testing

STATIC TESTING

- › What Is Static Testing?
- › Review
- › Inspection
- › Audit
- › Walkthrough

SDLC

- What Is Dynamic Testing?
- Types Of Dynamic Testing
 - Unit Testing
 - Module Testing
 - Integration Testing
 - System Testing
 - User Acceptance Testing

SOFTWARE PROJECT ENVIRONMENT

- Standalone Environment
- Client/Server Environment
- Web Environment
- Distributed Environment

TESTING LIFE CYCLE & IMPLETATION OF REAL TIME PROJECT

- Requirement Review
- Test Plan Preparation

- Test Strategy Preparation
- Test Case Identification

BLACK BOX TEST DESIGN TECHNIQUES

- ECP
- BVA

WHITE BOX TEST DESIGN TECHNIQUES

- Basis Path Testing
- Cyclomatic Testing
- Condition Testing
- Code Coverage

USE CASES

- What Is Use Case?
- UCD
- Identification Of Use Case Based Test Cases

TEST CASE DESIGN

- Introduction
- Test Case Design Template
- Types Of Test Cases
- Test Case Design Reviews
- Requirement Traceability Matrix
- Test Data Setup

TEST DATA

- What Is Test Data?
- Importance Of Test Data In Testing
- Approach For Gathering Test Data
- Benefit Of Test Data

TEST EXECUTION

- Introduction To Test Execution
- Test Execution Life Cycle
- Retesting

- Regression Testing
- Database Testing

RESULT ANALYSIS

- What Is Result Analysis?
- Comparing Actual Result Vs Expected Result
- Status Of Result

DEFECT TRACKING USING JIRA

- What Is Defect?
- Categories Of Defect
- Preparation Of DPD (Defect Profile Document)
- Defect Matrix
- Defect Reporting
- Defect Life Cycle Process
- Defect Removal Efficiency

TEST MEASUREMENT TECHNIQUES & METRIC COLLECTION

- Importance Of Test Metric
- Testing Data Used For Metrics
- Different Types Of Metrics

TEST CLOSURE

- Test Execution Stop Criteria
- Test Summary Report Preparation
- Status Of Result

USER ACCEPTANCE TESTING(UAT)

- What Is UAT?
- Importance Of UAT
- Scope Of UAT
- Alpha Testing
- Beta Testing

TESTING TECHNIQUES IN LIVE ENVIRONMENT

- Smoke Testing
- Sanity Testing
- Re-Testing
- Regression Testing
- Static & Dynamic Testing
- Deployment/System Testing
- Probability/Compatibility Testing
- Usability Testing
- Exploration Testing
- White Box, Black Box & Gray Box Testing
- Forced Error Testing
- End-To-End Testing
- Security/Penetration Testing
- Scalability Testing
- Reliability Testing
- Accessibility Testing

- Ad-hoc Testing
- Agile Testing
- Informal/Big-bang Testing
- Incremental Testing/Formal Testing
- Localization Testing
- Performance Testing
- Load & Stress Testing
- Recovery Testing

TEST MANAGEMENT PROCESS

- What Is Test Management Process?
- Requirements
- Business Components
- Test Plan
- Test Lab
- Defects

SITE ADMINISTRATION

- › Creating Domain/Project
- › Adding Users To Project
- › Creating Groups
- › Customization Of Quality Center, Release & Cycle Creation
- › Test Requirements
- › Example Of Test Requirements
- › Importance Of Tracing & Tracking Requirements
- › Reviewing & Building Requirement Structure
- › Entering Requirement Manually

TEST CASE CREATION & MANAGEMENT

- › Review Of Existing Test Case Parameter
- › Requirements
- › Business Components
- › Test Plan
- › Test Lab
- › Defects

TEST SET & TEST EXECUTION

- Creating Folder & Test Sets
- Defining Test Execution Flow
- Setting Test Set Properties
- Manual Test Execution
- Logging Defects During Manual Testing
- Automated Test Execution
- Adding Test Hosts
- Running Test Set
- Setting Run Times

DEFECT TRACKING USING JIRA

- Reporting Defects
- Searching For Similar Defects
- Using Grid Filters
- Deleting Defects

REPORTING & ANALYSIS

- › Analysis Menu Graphs & Results
- › Creating Editable Reports
- › Reporting

BUG TRACKING TOOLS

- › Jira
- › Microsoft Azure

AUTOMATION TESTING

- › What Is Automation Testing?
- › Benefits Of Automation Testing
- › Manual Testing Vs Automation Testing
- › Types Of Automation Testing Tools
- › Tool Selection Criteria
- › Automation Testing Process
- › When Should Automation Testing Used & Avoided?

COMMENTS

- Purpose Of Comments
- Types Of Comments
- Comments & Shortcut Keys

JAVA VARIABLES

- Declaring Variables
- Option Explicit
- Naming Restriction For Variables
- Scope Of Variables
- Life Time Of Variables
- Assigning Values To Variables
- Scalar Variables & Array Variables
- Dynamic Arrays

JAVA DATA OPERATORS

- Operator Precedence
- Arithmetic Operator
- Comparison Operator
- Logical Operator
- String Manipulation

INPUT/OUTPUT OPERATIONS

- Input Box Function
- Msg Box Function

CONDITIONAL STATEMENT

- If Statement
- If...Then...Else Statement
- If...Then...Elseif Statement
- Select Case Statement

LOOPS

- Do Loop
- Do Until Loop
- While Loop
- For Loop
- For...Next Loop
- For...Each...Next Loop

ERROR HANDLING

- Handling Syntax & Run Time Error
- Coding To Avoid Error
- Coding To Handle Error
- The Error Object
- The Raise Method

BUILT-IN FUNCTION

- String Function
- Math Function
- Date & Time Function

DEBUGGING

- What Is Debug?
- Types Of Bug
- Steeping Through Code
- Watching Variable Values

OBJECT

- Introduction To Object
- What Are Objects?
- Properties Defined
- Methods Defined
- Event Defined
- Referencing Object

OBJECT REPOSITORY

- Introduction
- How Objects Are Added To Object Repository
- Identifying Object
- Viewing Object Properties
- Object Repository Modes
- Per-Action Vs. Shared Object Repository
- Object Spy
- OI Introduction
- User Defined Object
- Smart Identification
- Managing Object Repository
- Merging Shared Object Repositories
- Associating Shared Object Repository To
- Test
- Merging Object Repositories

SYNCHRONIZATION

- › What Is Synchronization?
- › Synchronizing Your Test
- › Synchronization Methods
 - Adding Wait Method
 - Adding Wait Property Method
 - Adding Exist Method
 - Adding Sync Method
- › Global Synchronization Settings
- › Browsers Vs. Page Sync
- › Web Event Recording Configuration

CHECKPOINTS

- › Introduction
- › Adding Checkpoints While Recording
- › Types Of Checkpoints
 - Text Checkpoint
 - Text Area Checkpoint
 - Bitmap Checkpoint

- Database Checkpoint
- Accessibility Checkpoint
- XML From Recourse Checkpoint
- XML From Application Checkpoint
- Page Checkpoint
- Table Checkpoint
- Image Checkpoint

➤ Updating Checkpoints At Run Time

OUTPUT VALUES

- Introduction
- Types Of Output Values
- Creating Output Values
- Storing Output Values

ACTIONS

- Introduction To Action
- Types Of Actions
- Creating Actions
- Inserting Call To Actions

- › Insert Call To Copy Of Action
- › Insert Call To Existing Action
- › Splitting Actions
- › Input And Output Parameters
- › Action Template

PARAMETERIZATION TEST & DATA DRIVEN TESTS

- › What Is Parameterization?
- › Parameterizing Using Actions
- › Parameterizing Using Data Tables
- › Parameterizing Using Environment Variables
- › Build-In Environment Variables
- › Internal Environment Variables

WORKING WITH DATA TABLES

- › Introduction
- › Working With Global And Action Sheets
- › Editing And Saving Data Tables

- Data Table Methods
- Data Tables And Database
- Importing & Exporting Data To /From Spread Sheet

REGULAR EXPRESSIONS

- Introduction To Regular Expression
- Defining Regular Expression
- Running And Analyzing Expression
- Regular Expression Character
- When To Use Regular Expressions In Quick Test

FUNCTIONS

- Introduction
- Creating User Defined Functions
- Advantages If Using Functions
- Creating Generic Functions

- Function Decomposition
- Read From Excel
- Write To Excel
- Step Generator

RECOVERY SCENARIOS

- Introduction To Trigger Events
- When To Use Recovery Scenarios?
- Default Recovery Scenarios
- Recovery Function Call
- Recovery Scenarios Manager
- Error Handling
- Creating Recovery Files
- Associating Recovery Files

VIRTUAL OBJECTS

- Introduction
- What Is Virtual Object?
- Creating Virtual Object

- Virtual Object Manager
- Entire Parent Hierarchy
- Parent Only

MAPPING USER-DEFINED TEST OBJECT

- Introduction
- Object Mapping
- Standard Classes

DEBUGGING IN QUICK TEST

- Establishing Our Debugging Configuration
- Using Break Points
- Step Into, Step Out And Step Over
- Debug Viewer Pane
- Working With Watch Expressions Tab
- Working With The Variables Tab
- Working With The Command Tab

LIBRARY FILES

- Associating A Library Globally With Test
- Dynamically Loading Libraries Locally At Run Time
- Dynamically Loading Libraries Globally At Run Time
- Applicability Of Option Explicit

WORKING WITH DATABASES

- Introduction
- Checking The State Of Connection Or Record Set
- MS Access, SQL Server And Oracle
- Exporting The Results Of A Query To Data Table
- Database Testing Using SQL Server
- Database Testing Using CRUD Operations on Database

DEBUGGING IN QUICK TEST

- Establishing Our Debugging Configuration
- Using Break Points
- Step Into, Step Out And Step Over
- Debug Viewer Pane
- Working With Watch Expressions Tab
- Working With The Variables Tab
- Working With The Command Tab

WORKING WITH FLAT FILES

- Introduction
- Types Flat Files
- Modes Of Flat Files
- Advantages Of FSO

UTILITY OBJECTS

- The Crypt Object
- The Optional Step Object
- The Path Finder Object
- The System Util Object
- The Register User Function Object
- The Services Object With-End
With Transactional Statements

QTP AUTOMATION OBJECT MODEL

- Introduction
- Automation Quick Test Operations
- How Can We Use AOM To Launch
Quick Test On Remote Machine

WORKING WITH WEB TABLES

- Introduction
- Accessing The Web Tables
- Clicking Inside The Web Table
- Row Count
- Column Count
- Get Cell Data

WORKING WITH DATA TABLES (EXCEL)

- Introduction
- The Excel Automation Object Model
- The Word Automation Object Model

DESCRIPTIVE PROGRAMMING

- Introduction
- Advantages Of Descriptive Programming
- When To Use Descriptive Programming?
- Descriptive Programming Syntax

- Using Description Objects
- The MIC Class Identifier
- Converting Or Based Script To A DP-Based Script
- Types DP/PD Methods

IMPLEMENTING TEST AUTOMATION FRAMEWORKS IN REAL TIME PROJECTS

- Introduction
- Design Guidelines
- Advantages Of Frame Work
- Types Of Automation Frameworks
- Support Different Application Versions
- Implementing In Real Time Environment
- Disadvantages Of Frameworks

BROWSER AUTOMATION TESTING TOOL SELENIUM

- What Is Automation Testing?
- When It Comes Into The Picture?
- What Are The Uses Of Automation Testing?

INTRODUCTION TO SELENIUM

- What Is Selenium?
- Use Of Selenium
- When Selenium Can Be Useful In Testing
- Features Of Selenium
- Difference Between Selenium And QTP

SELENIUM COMPONENTS

- Selenium IDE
- Selenium RC
- Selenium Grid

API TESTING USING THE POSTMAN TOOL

- Understanding APIs and their role in software applications
- Types of APIs: REST, SOAP, GraphQL, etc.
- Key concepts: Endpoints, Requests, Responses, Headers, Payloads, Status Codes
- Tools for API Testing (e.g., Postman, Swagger, SoapUI, REST Assured)
- Writing and executing test cases for API validation
- Performing CRUD operations (GET, POST, PUT, DELETE)
- Testing authentication and authorization
- Validating response formats (JSON, XML)
- Common challenges in API Testing and best practices
- Real-world API testing scenarios and hands-on practice using Postman

LIVE PROJECT

- 1 for Manual Testing
- 1 for Automation Testing

We Will Learn Automation Projects Using
BDD Framework(cucumber)

Our Placement Process



Our Placement Companies



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